

TORSVAG FYR, Norway

WMO: 010330

Lat: 70.246N

Lon: 19.500E

Elev: 24

StdP: 101.04

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.8	-5.7	-14.3	1.1	-5.1	-12.4	1.3	-4.4	19.9	0.1	18.1	0.2	7.9	120	0.802

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	18.7	13.5	17.1	12.5	15.7	11.9	14.2	17.8	13.1	16.2	12.2	14.9	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.4	9.0	15.9	11.4	8.4	14.6	10.6	8.0	13.6	40.1	17.9	37.2	16.3	34.8	15.2	17.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.3	12.6	DB	-8.5	21.4	1.9	1.7	-9.8	22.7	-10.9	23.7	-12.0	24.6	-13.4	25.9	(4)
(5)			WB	-10.0	15.6	2.0	1.2	-11.4	16.5	-12.6	17.1	-13.7	17.8	-15.2	18.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	0.0	-0.3	0.1	2.5	5.9	8.5	11.3	11.2	9.2	5.6	2.9	1.3	(6)	
	DBStd	5.04	2.94	2.89	2.73	2.92	3.04	2.64	2.67	2.46	2.78	2.89	2.75	2.86	(7)	
	HDD10.0	2023	311	289	308	225	132	61	14	14	46	139	213	270	(8)	
	HDD18.3	4912	569	522	566	474	385	296	218	223	273	395	463	528	(9)	
	CDD10.0	153	0	0	0	1	6	16	56	50	23	3	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)	
	CDH23.3	1	0	0	0	0	0	1	1	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	5.6	6.8	7.0	6.9	5.4	4.9	4.3	3.6	5.0	5.9	6.4	7.2	(14)	
(15) Precipitation	PrecAvg	821	74	61	75	54	51	44	49	58	87	108	77	83	(15)	
	PrecMax	1023	176	120	150	148	136	122	135	135	167	222	156	140	(16)	
	PrecMin	590	0	14	17	10	10	16	2	9	13	15	0	0	(17)	
	PrecStd	113	46	25	34	29	31	23	34	33	40	43	43	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	6.7	7.5	12.3	17.0	19.0	22.0	19.7	17.4	14.0	10.1	8.3	(19)	
		MCWB	4.7	4.1	3.5	7.0	10.5	13.0	15.1	14.1	11.9	9.7	7.0	5.6	(20)	
	2%	DB	5.9	5.7	6.1	9.5	14.3	16.5	19.7	18.0	16.0	12.1	8.9	7.2	(21)	
		MCWB	3.6	3.4	3.6	5.9	9.2	11.6	14.2	13.4	11.5	8.7	6.6	5.1	(22)	
	5%	DB	5.0	5.0	5.1	7.5	12.2	14.4	17.5	16.5	14.4	10.5	7.9	6.3	(23)	
		MCWB	2.9	2.9	2.8	4.6	8.0	10.6	13.0	12.6	10.8	7.9	5.7	4.1	(24)	
	10%	DB	3.9	3.9	4.0	6.3	10.3	12.5	15.7	15.0	12.9	9.4	6.8	5.3	(25)	
		MCWB	2.0	2.0	2.0	3.9	6.8	9.5	12.4	11.9	9.9	7.3	4.7	3.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	5.1	4.7	7.5	11.1	14.0	16.1	15.4	12.7	10.2	8.3	6.5	(27)	
		MCDB	6.5	6.3	6.0	11.5	15.9	17.6	21.2	18.7	15.8	13.1	9.5	7.7	(28)	
	2%	WB	3.9	3.7	3.9	6.1	9.5	12.0	14.7	13.9	11.9	9.2	7.0	5.4	(29)	
		MCDB	5.6	5.1	5.6	8.8	13.8	15.6	18.5	17.1	15.1	11.3	8.6	7.0	(30)	
	5%	WB	3.0	2.9	3.0	5.1	8.3	10.8	13.6	13.0	11.1	8.3	5.8	4.3	(31)	
		MCDB	4.7	4.4	4.9	7.2	11.7	13.8	16.9	15.7	13.6	10.2	7.6	6.1	(32)	
	10%	WB	2.2	2.0	2.1	4.2	7.2	9.7	12.5	12.1	10.4	7.4	4.8	3.2	(33)	
		MCDB	3.8	3.7	3.8	6.1	9.6	12.1	15.2	14.5	12.6	9.3	6.6	5.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.7	2.9	2.8	3.4	3.9	3.8	4.2	3.7	3.3	2.7	2.6	2.8	(35)	
		MCDBR	3.7	3.8	3.5	4.6	6.9	7.2	8.0	6.5	5.3	3.7	3.8	3.9	(36)	
	5% WB	MCWBR	3.3	3.2	2.8	2.9	3.9	4.1	4.4	3.6	2.9	2.7	3.2	3.4	(37)	
		MCWBR	3.5	3.8	3.5	4.5	6.5	6.5	7.5	5.9	4.8	3.5	3.7	4.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.271	0.322	0.335	0.344	0.341	0.346	0.335	0.317	0.272	N/A	N/A	(40)	
		taud	N/A	2.097	2.199	2.334	2.438	2.498	2.516	2.529	2.522	2.188	N/A	N/A	(41)	
	Ebn at Noon		0	423	663	796	839	854	834	801	706	416	0	0	(42)	
		Edh at Noon	0	39	73	88	90	88	84	73	55	34	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.22	1.17	2.93	4.47	4.89	4.26	3.00	1.59	0.45	0.00	0.00	0.00	(44)	
	RadStd	0.00	0.04	0.14	0.30	0.34	0.51	0.45	0.23	0.18	0.07	0.00	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page